

Communication Systems
Weekly-Test-VI
Discussion Forum Topics
16 March 2026 to 20 March 2026

1. Why is FM radio preferred over AM radio for music broadcasting? Explain with reasons.
2. Where do you see analog communication systems still being used in everyday life today? Give examples.
3. Explain how a traditional radio receiver works using concepts from analog communication.
4. How does noise affect communication signals in real life? Provide an example.
5. Why does radio signal quality reduce when moving away from the transmitter?
6. Explain the role of antennas in analog communication systems.

7. Why is modulation necessary in communication systems?
8. What would happen if communication systems transmitted baseband signals directly without modulation?
9. Compare Amplitude Modulation (AM) and Frequency Modulation (FM). Which one is better and why?
10. Why does FM provide better noise immunity than AM?
11. Explain the concept of bandwidth using a real-life example.
12. Why is signal-to-noise ratio important in communication systems?

13. If AM uses more power for the carrier signal, why is it still used today?
14. What are the limitations of analog communication systems compared to digital systems?
15. Why do engineers prefer frequency modulation for high-fidelity audio transmission?
16. What challenges occur when transmitting analog signals over long distances?
17. Why is filtering important in communication receivers?
18. How does interference affect communication systems in crowded frequency bands?

19. Imagine you are designing a communication system for a rural area. Would you choose AM or FM? Explain your choice.
20. If you were designing a radio station, what factors would you consider for signal transmission?
21. Suppose a radio signal has severe noise. What techniques could improve the signal quality?
22. How would you explain modulation to someone with no engineering background?
23. If communication systems did not use modulation, what problems might occur?

24. Even though digital communication is popular today, why is it still important to study analog communication?
25. How does analog communication form the foundation for modern wireless technologies?
26. Can analog communication systems be useful during emergency communication situations? Explain.
27. What role does analog communication play in broadcasting industries today?

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28. Describe a real-life communication system and identify its main components (transmitter, channel, receiver).
29. Explain modulation using a creative analogy (for example: music, traffic, language, etc.).
30. If you were teaching analog communication to school students, how would you explain it in a simple way?